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Introduction

The NHS Counter Fraud Authority (NHSCFA) for years had a strategic priority to examine fraud within general practice (GP), aiming to identify areas in England who have irregular patterns of patients. Yet this objective has never been complete, remaining a constant waste on valuable resource.

GP's on average receive £140 per patient to provide unlimited care, with 60.5 million patients registered with a GP, means the NHS spends 9.4 billion on care in the community.

Fraud estimates in this area are costing the NHS £88 million (NHSSIA, 2018), but the organisation has no planned approach of how to tackle this objective.

Product Design

Data source and theme selection and specification

Therefore, as an Information Specialist and project within the NHSCFA, it is their responsibility to show how the value of open-source data can provide direction against the unmet priority, seeking to enhance the organisations antiquated manual methods of intelligence gathering by making data accessible and applying context to the problem resulting in enhanced perception through the use of visual means.

The product developed will follow the data management lifecycle (Cady, 2017) resulting in the production of a self-service tool empowering the audience (domain expert) to go on a visual journey, with Hilde et al. (2019) conveying the importance of drawing on the audience's expertise and positioning them at the centre of the analysis process, aiding every user to learn new analytical skills to detect anomalies, guiding the experts into making their own business decisions using data which previously didn't mean much on its own.

As the intended audience is seeking to detect anomalies then engagement with experts who are the intended audience have formed the following interestingness measurements (McGarry 2019), focusing on the following anomaly points;

1. A data point will be classified as an anomaly if the number of people on a GP lists is more than those register on the Census at various dimensions (Country and CCG level)

2. Difference by Age group for the 65+ age range (Census V CCG)

3. Any Clusters of behaviour for organisations which differ from the norm.

The product will be built using 3 open-source structured raw data files containing all unclassified information, identified by NHSCFA with variable selection complete when gathering requirements, (see appendix 1 for links and location of transformed data). Confirming, Ma (2019) argument that it's easier to access open-source data to tackle the problem first, though Nunnes (2020) argued this can be restricting giving the detailed information required to prove fraud.

No integrity issues were identified as the sources are reliable. Yet data assurance must be evaluated following the extract transform and load (ETL) stage, often problems are generated

when wrangling the previously raw data e.g. duplicates. Critically from the offset, the project lead must understand the limitations of the raw data.

The time frame was dictated by the Census dataset due to its infrequent publication of the net-migration calculation. Therefore, all datasets have been gathered for the timescale 2016 to 2018 except when interacting at CCG level, due to the capabilities of the free software. Yet this may confuse the audience causing clear titles detailing timescale to be incorporated.

Application domain/end user's requirements analysis

Domain experts within the Intelligence arm of the NHSCFA will be the end-user detailing requirements including the ability to use simple calculations to visually detect anomalies. Currently, computational methods are viewed unreliable, meaning by immersing the audience at the forefront of the analysis process, will provide them with a trustworthy platform to make confident decisions.

The measures selected will focus on the use of descriptive statistics, e.g. calculating difference between variables, e.g. Census versus GP-List/CCG and by age category.

Additionally, the user although not analytical, requested the use of unsupervised clustering methods, seeking to identify groups of behaviour which would usually be invisible. Though as experts are non-technical then Amarasinghe et al. (2018) argued it is pivotal the above requirements are visually appealing and easy to use.

Principles of Visualisation

Critically the product must be applicable to the problem, enticing the audience to exploit, reaping rewards from the product presented and decisions formed, therefore three principles (see figure 1) must be demonstrated by the visual product.

Figure 1: Three Principles of Good Visualisation Design



Source: Kirk (2016)

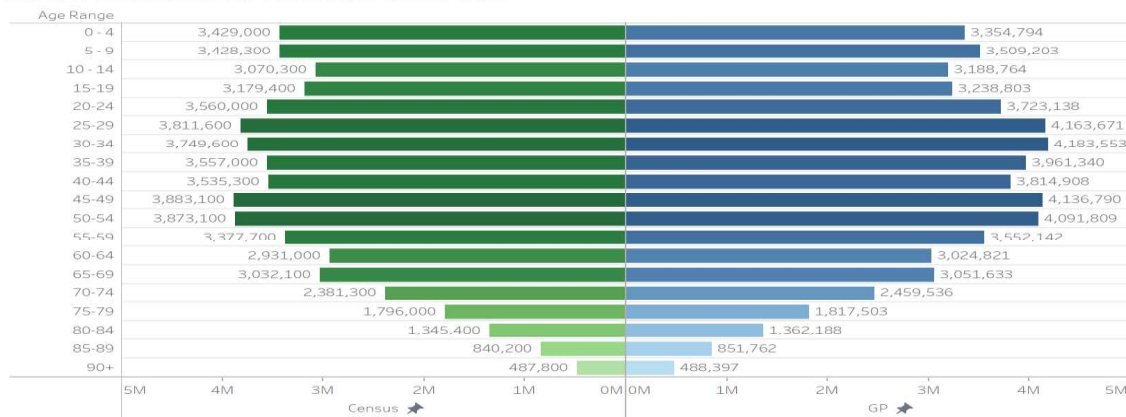
These steps must be constructed alongside the experts, with the information specialist (project lead) demonstrating options. Yet as curator of the data and creator of its portrayal (Kirk, 2016, p.36) then unwanted bias can introduce subliminally into its editorial content.

Therefore, consideration must be given to the colour design. The NHS commonly applies hues of blue/orange and is reluctant to use the red colour pallet due to its signification of an emergency (NHS-England, 2020).

To illustrate patterns which are not apparent and after engagement the use of the green/blue hue pallet will be applied to indicate the difference between the Census and GP sources, appointing green for Census variables considered to be the hue associated to 'land' and blue disguising the NHS (demonstrated in figure 2).

Figure 2: Use of Colour pallet

Age Range Comparison (2016 to 2018)
Number of people registered on Census against those on GP lists



Visual terminology involving measurement calculated will use a gradient hue to visually display difference in the charts or mapping presented. The only time a red hue pallet will be applied is to identify individual loss per CCG on Dashboard 2, shown in figure 3 below and on the Density Based clustering to signify higher values.

Figure 3: Dashboard 2 - Loss value visual representation.



Product functional and non-functional requirements specifications

After discussion with the domain experts, the below requirements in table 1 were developed.

Table 1: Product Functional and Non-Functional Requirements

Functional	Non-Functional
Comparison of Trend GP v Census (to compare values easily)	User-friendly interface, with minimal instructions on how to use (e.g. Filters)
Use of difference coloured hues to distinguish variance where the proportion of people registered on GP list is over the Census population	Enhance Visual Perception by visually representing all data sources
Interactivity - Filters for drilling down into anomalies and data segments	Immerse the domain specialist into the centre of the analysis process

Outlier detection using clustering methods	Responsive software platform allowing audience members to react and progress through the story quickly
Geospatial plotting by lowest dimension within data e.g. (CCG) – to illustrate patterns that are not immediately apparent aiming enhance the users' visual perception	Free up resource time, reducing operating costs by using available data, instead of conducting manual antiquated methods
Extract drill down results into PDF	

Source: Conversation with Wyke and Hipworth (2020)

Product software architecture design

The architecture for the product will be a client-server model designed to connect different data sources securely (See figure 4). Although the organisation employs SAS product this is restricted, supporting Smith (2017) insight of BI system purchased on false promises.

The data server is the first layer in the architecture which connects data in various heterogeneous environments (Tableau-blueprint, 2020). Secondly is the connector layer, connecting databases using its ODBC connector. Tableau is then used to connect to the database either locally or utilising its web servers.

The third layer in the architecture is the visualisation product tableau desktop (version-2020.1), using the web-based platform to allow the visualiser/creator to produce dashboards and visualisations in Tableau server which can then be viewed by the Intelligence audience experts.

Figure 4: Tableau Product Architecture

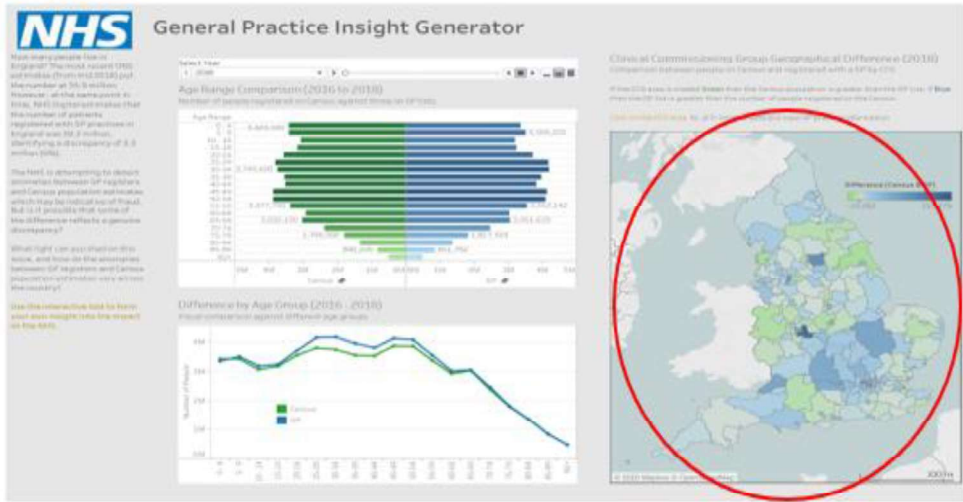


As the product requires connection to Tableau server via the internet and as the organisation's internet server is based in London then a risk involving bandwidth performance especially currently where the entire organisation is connecting from home. To circumvent this problem, it's advised that virtual machines are assembled to bypass the bandwidth issue connecting directly to source.

Product user case specifications

The non-technical user has highlighted the following elements to be an integral part of the product, including:



Data Visualisation	Extract insight and allow the user to become fully immersed in the story of the data. This will be achieved by using filters, colours to highlight and differentiate between the different dimensions within each data source, guiding though not influencing the audience member. Thought to the colour hue used to represent the two different data sources, as well as the positioning of the dashboard element, was undertaking. The user requested that each dashboard presented be visible without scrolling, causing several actions and prompts to be built into the product. Interaction
Animation	Where data is presented over time than the ability to aminate the time sequence to alter visual perception.
Geospatial	Representation of variance using a choropleth mapping element highlighted below and subsequent drill down using cluster mapping allowing the data to become 'alive' to the user cognitive process. 
Value	New Measure to be created showing monetary value of difference between GP and Census.

To ensure the project lead produces quality relevant work, then the following questions (see Appendix 2) must be reflected on frequently.

Product Development

The selection of appropriate software tools/platforms and hardware methodologies

The selection for the computer platform was anchored on the audience members skills level and how the ecosystem overall could contribute and strengthen the organisation's ability to answer the business problem posed alongside 3 key project management principles including is the platform for developed and distribution, fast, good and cheap.

Considering these factors, the tools utilised by the data engineer (project lead) include R-studio and R-coding language (Version-4.0.0) in the ETL phase, followed by extract into CSV format. No audience members were involved in this stage.

Next via the trusted and governed Tableau platform, data is visualised as per user requirements and specifications. Additionally, to compliment the Tableau platform, Rserve will be utilised acting as a socket server (TCP/IP or local sockets) allowing for requests to be sent to R.

Attention to the laptop device configuration setting used by the audience to access the product must be tested. However, the organisation has recently upgraded hardware with many now having the functionality of switching to tablet mode, thus automatic display settings have been configured on the tableau desktop product before publishing to the public server.

Product development software engineering methodology

To create and developed the product the Waterfall software engineering methodology was initially selected, but due to restriction by some of the audience then this developed into an incremental approach, reiterating El-Masri (2013) notion that the methodology used should be based on the problem and environment.

This approach was employed to gain value earlier in the development process, incorporating feedback from the selected audience member and amending the system to meet the user's requirements particularly regarding visual aspect of the geographical mapping capabilities.

System testing method

The testing phase built into the incremental approach involved two audience members (domain expert) who tested the product operates as expected. The user tested all functional and non-functional requirements such as the drilling into a lower dimension of the data. User feedback was then be incorporated into stage 2 development

User evaluation plan and methods

Evaluation was conducted against two stages of development. First, the accuracy of the cluster model on dashboard 3 and the human/computer integration between the functionality and integration of visual elements developed to motivate use.

Clustering:

Goldstein and Uchida (2016) conveyed successful results detecting anomalies with K-means though Domingues (2018) identified an issue when selecting the number of clusters to apply. Therefore, due to the ambiguity, the NBCLust package in R was run against the GP List Sizes. The result found 3 clusters to be the optimum as displayed in figure 5. However, when examining the scree plot (See figure 6) the elbow-point where the eigenvalues level-off was found to indicate 8 clusters to be significant, confirmed correct by the domain specialist.

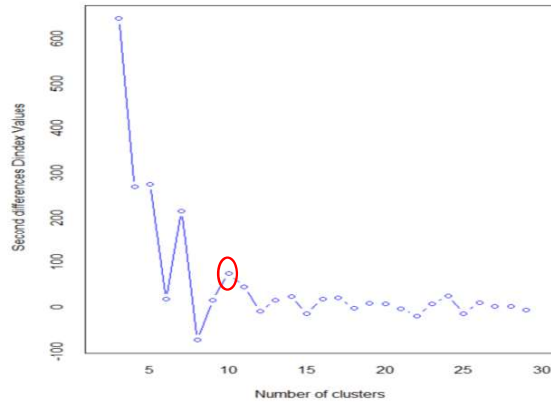
Figure 5: NBCLust Results

```
*****
* Among all indices:
* 1 proposed 2 as the best number of clusters
* 2 proposed 3 as the best number of clusters
* 1 proposed 14 as the best number of clusters
* 1 proposed 23 as the best number of clusters
* 1 proposed 26 as the best number of clusters

***** Conclusion *****

* According to the majority rule, the best number of clusters is 3
```

Figure 6: Scree Plot



To ensure accuracy both default algorithms in R and Tableau were evaluated. Finding R to have an accuracy measure of 96.4% (figure 7) compared to Lloyds 95.8% in Tableau (figure 8). However, due to the negligible difference, the domain experts evaluated Tableau's as acceptable.

Figure 7: R (Hartigan and Wong) Model Accuracy

```
within cluster sum of squares by cluster:
[1] 5143950816 5684954826 4888547232 5596486935 3209372362 4818825574 5383292831 5482044522
(between_SS / total_SS = 96.4 %)
```

Figure 8: Tableau (Lloyds) Model Accuracy

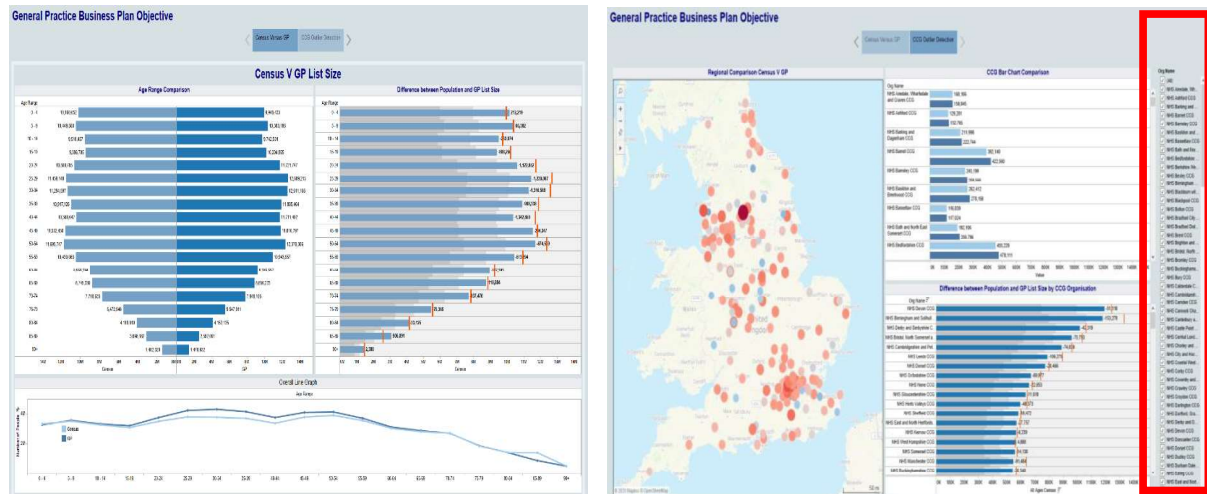
```
within cluster sum of squares by cluster:
[1] 10892405220 5272036042 140261808617 2300845748 1549376401 19386460764 28190275889 104750967807
(between_SS / total_SS = 95.8 %)
```

User Experience:

Although unplanned the initial product developed was released to two users for feedback with advice used to drive redevelopment.

However, when presenting the first prototype for testing major changes were sought, alternating 80% of the product. Initially the product was spread across two storybooks displayed in figure 9 below, but upon testing, tableau functionally didn't allow for links between books, meaning dashboard actions had to be created to meet the specification.

Figure 9: Prototype 1 delivered to audience



Also, the audience disliked the list of CCG's (highlighted in red-figure 9) but instead requested drilldown into individual CCG's calculating the difference in lists, causing the final product to include search facilities and annotation containing CCG information.

Although the audience liked the interaction between dashboards, a method for switching between specific tabs was requested, causing for the worksheets to appear visible as a row header. The project lead advised against this given the non-technical ability the confusion this could bring but was overridden. With the final visual output displayed in figure 10,11 and 12 below)

Figure 10: Homepage – Dashboard 1 (Animation on age and drill down map)

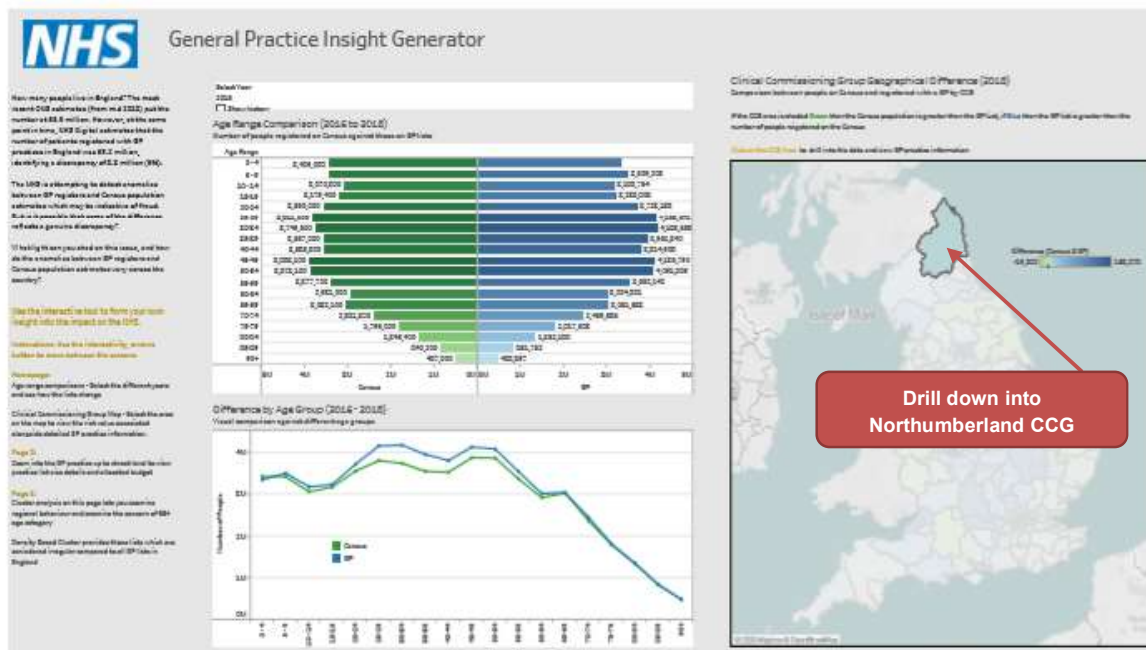


Figure11: Page 2 (Drill down CCG risk value and GP details)

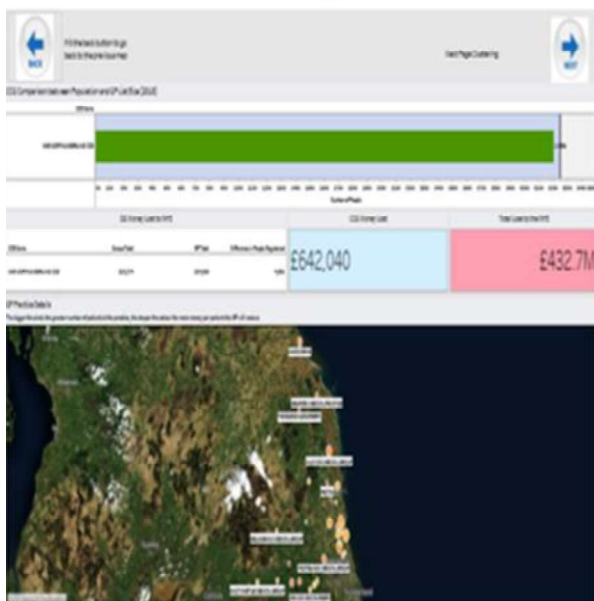
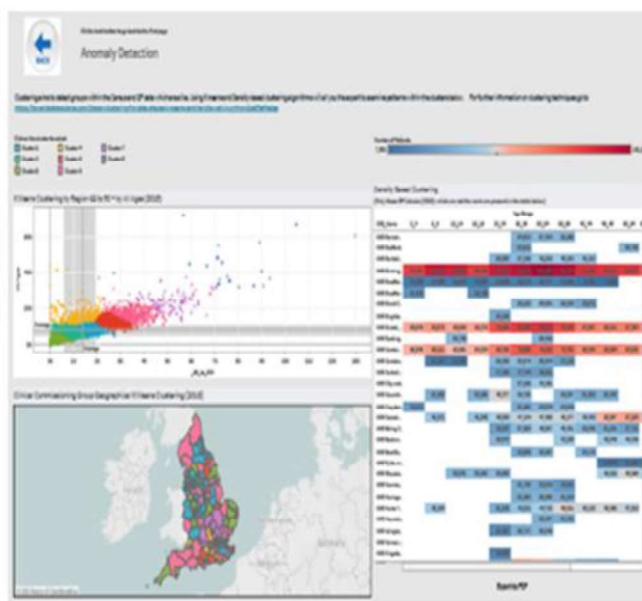


Figure 12: Page 3 (Anomaly Detection)



Published product can be found at:

<https://public.tableau.com/profile/susan1454#!/vizhome/Assignment2NHSAnomalyInsightGeneratorFinal18May2020888/HomePageDashboardD1?publish=yes>

Project Management

Time Management – Gantt Chart

To ensure the project is controlled then task duration and dependencies have been illustrated in a Gantt chart (see Appendix 3) permitting the project team to view each task and resource alongside if the tasks/activities are on time or require additional intervention.

The project was in development at a time of stress upon the NHS meaning engagement which previously complete face-to-face was executed using digital methods, with scheduling constraints built into the project plan.

Risk assessment on personal information protection and data security/governance

All data utilised is open-sourced extracted from reliable Government platforms, this aspect played an important part when selecting the distribution platform.

Any decisions made using the product will only cite the source and methods which influenced the decision. Though as the NHSCFA main ethical principles is to act with public interest at the forefront of any business decision (IEEE) is risk-averse due to the sensitive nature of the subject, then an ISO risk standard of level 4 would be applied to any anomaly detected, with subsequent investigation commenced before publication of the insight.

Notably, if anomalies then become criminal investigations then the master data management of the product must follow best practice guidelines, documenting sources from which the product is created (Bitterer et al. 2012).

Quality control on software development

To ensure the quality of the product developed is in line with user specifications and requirements then all features and functionality of the product were checked by the project lead.

Expected behaviour was then tested using several test examples designed by the audience, with several problems encountered causing additional actions to be built into the product. Also, as the product displays statistical calculations computed within Tableau then supplementary quality assurance (QA) tests against all dashboard outputs were undertaken. This ensured integrity of the product while installing confidence to the audience.

It must be noted the audience members also highlighted duplicate issues within the beta version (prototype-1), caused in the ETL stage, causing additional QA features being coded into R before any transformed dataset could be uploaded into Tableau.

Basic Customer/User relationship management

The audience was engaged from the initial conversation leading to the development of the visual product. Their needs were explored and discussed from a practical user perspective, considering their non-technical ability.

However, the product was still developed utilising SMART methodology so clear boundaries and definitive endpoints were established. However, due to Covid-19 restriction, all engagement took place by digital format e.g. Windows Teams and email.

After engagement of the initial product that was designed as per the original specification, amendments were released once the audience had time to evaluate the beta product. This caused costs associated with resource time to increase and be factored into the timescale.

Where specific changes were requested but beyond the scope of the product specifications or platform these were explained to the requestor as unfeasible at initial live release but considered for future development.

Basic Product marketing strategy

The product marketing approach involved demonstrating the use to audience members by answering the real-life business problem, e.g. organisational hypothesis is the problem centres within age group 65-plus as GP's receive more money per patient as age increases. Yet the interactive product has uncovered this not the case, finding the prevalent pattern to be in working age. Demonstrating this insight alone will showcase new approaches to tackling problems while developing cognitive responses using data.

Conclusion

The project demonstrated that having a real-life business problem and using open source data to harness wisdom can effectively answer once previously unobtainable insight simply by developing an impactful visual data product. However, this product can't be developed in isolation and must include stakeholder engagement from the offset, though all aspects of the product must be realistic otherwise expectation will not be met and the product fails.

However, in an organisation which is reluctant to utilise any form of digital data, preferring to use antiquate methods then this developed product will provide the domain expert with the visual superpowers to find new anomalies previously hidden from sight. Helping not only the

domain expert to facilitate understanding by being a master of their own decisions but permits the NHSCFA to achieve its objective.

Recommendation

Lesson gained from the project includes future utilisation of the Agile project methodology from the offset to facilitate any change within the working environment. Besides, face to face engagement with the audience of the initial product when demonstrating all features and dimensions, with consideration to the audience members selected for testing. Though as the infrastructure while working from home is underload then consideration of the BYOD including the mobile platform must be examined for any future development phases.

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